

3. **Virtualization for Directed I/O (VT-d):** If you read the Fast Track to Virtualization we published last year, you're probably well acquainted with how storage is handled in virtualized environments. This technology helps improve the performance of a virtual machine when it tries to access data on disk. The overall concept behind VT-d is hardware support for isolating and restricting device accesses to the owner of the partition managing the device.
4. **VT-x for Extended Page Tables:** This technology helps the virtualization software access memory for virtual machines and improves performance further.
5. **Thermal Monitoring Technology:** This helps reduce the power consumption as well as helps protect the hardware in case of over-heating. One useful add-on for sure!
6. **AES New Instructions:** BitLocker, the disk encryption technology introduced with Windows Vista and present in Windows 7 and Windows 8 uses AES algorithm for encryption/decryption. If you plan to use BitLocker on your laptop, this feature would improve performance by a noticeable margin. Besides this, you wouldn't find any other use for it (unless you're designing software which employs this feature).

Though what we just mentioned is in reference to laptop processors, (i.e. the regular bulky notebooks), the same stands true for ultrabooks as well as convertibles since they share similar hardware despite different form factors.

When opting for an ultrabook, ensure that it has proper cooling and a processor with Turbo Boost. While mobiles and tablets have lower processing requirements, we can't say the same for ultrabooks which take on more load. Depending on the load, the heat generated would rise. Lack of cooling could mean shorter battery and component life. Cooling, in most cases, is availed by ventilating holes at the bottom of the machine. Make sure that there are enough air vents before falling for looks.

GPU

All work and no play makes for a dull laptop

Hearing that a laptop comes with "1 GB of Graphics Card" is not uncommon. One scratch under the surface and you would find that almost all graphics cards come with 1 GB or more of memory. But is it just the '1GB' that matters? Obviously not. If it did, all laptops and all graphics cards would have been the same and the rest of the GPU specifications wouldn't have mattered at all.